

The empirical recurrence for OEIS A235253: recovering a conjecture that is not written down

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Abstract

OEIS A235253 records “Empirical recurrence of order 57”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 2744$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 57, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 57 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235253 is “Number of $(n+1) \times (3+1)$ 0..7 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 7, with no adjacent elements equal (constant-stress tilted 1×1 tilings).”. It has offset 1 and begins

2672, 8328, 24232, 80812, 251656, 878988, 2873128, 10362212, ...

The entry states:

Empirical recurrence of order 57 (see link above).

As of the “Last modified” line on the live entry (Jun 18 2022 23:51:58, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2744$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2744$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 5488$ exact terms of the model returns order 57 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	9	c_{16}	1119106172061	c_{31}	−162403541094144492	c_{46}	−854924723263660032
c_2	127	c_{17}	6184991138491	c_{32}	918481782201225388	c_{47}	−205609048845972480
c_3	−1351	c_{18}	−13876898350953	c_{33}	287494045203027008	c_{48}	290534449453498368
c_4	−6970	c_{19}	−51315982162855	c_{34}	−1863700974113722744	c_{49}	78863727975432192
c_5	94422	c_{20}	132856045277474	c_{35}	−422013928865754048	c_{50}	−70519170047901696
c_6	203199	c_{21}	341593943001294	c_{36}	3015540535083578768	c_{51}	−21230668399804416
c_7	−4081841	c_{22}	−1004168415842563	c_{37}	538265483103102304	c_{52}	11522220940984320
c_8	−2552238	c_{23}	−1827034278729575	c_{38}	−3868024236997155488	c_{53}	3783927548805120
c_9	122295118	c_{24}	6063696097946702	c_{39}	−621258543394255104	c_{54}	−1132380684288000
c_{10}	−38584669	c_{25}	7843973087820822	c_{40}	3904358519038918720	c_{55}	−399811313664000
c_{11}	−2696627305	c_{26}	−29432051415735201	c_{41}	642969817150991360	c_{56}	50451185664000
c_{12}	2615932827	c_{27}	−26963713207037289	c_{42}	−3072362471867924864	c_{57}	18919194624000
c_{13}	45358608629	c_{28}	115104474905035461	c_{43}	−562991183742548736		
c_{14}	−66629192318	c_{29}	74025300218642079	c_{44}	1861965011974725120		
c_{15}	−595315696508	c_{30}	−362637605346244798	c_{45}	391138963834897152		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 57, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $57 + 4$ terms removed returns order 57 as well, so $\deg q_{\min} = 57 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $57 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 57 that this sequence satisfies — and that family turns out to have exactly one member.

References

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